

Department of Computer Science and Engineering
B.Tech. 3rd Year | 5th Semester

Software Engineering | CSE3102
Assignment 1

Instructions:

1. All questions are compulsory.
2. Assignment must be handwritten.
3. Assignments will be submitted over LMS (scanned pdf of handwritten assignment) as well as in hard copies to respective instructors.
4. **Deadline for LMS Submission: 04th September 2026**
Deadline for Hard Copy Submission: 07th September 2026
5. No extensions will be entertained.

Question-1

A team is doing release planning, and they decide the next release will include all stories from Story 1 to Story 11 listed below.

- The velocity range to be used for release planning is 14–20 ideal days per iteration.
- The team works in a 2-week iteration.
- It costs \$48,000 per iteration to fund the entire team.

Calculate the estimated duration for the next release (in iterations and weeks). Additionally, how much will this release cost?

Prioritized Product Backlog (estimates in ideal days):

Story Title	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11
Estimate (in ideal days)	4	6	5	3	7	2	5	6	4	8	3

Question2:

Consider the following mobile banking application requirements:

- Login using portal account
- View account balance
- Deposit cheques
- Check offers and promotions
- Take pictures of cheques
- Enter cheque details
- Select account
- Verify image with details
- Print receipt

Create a User Story Map by organizing these requirements into:

- User Activities
- User Tasks
- User Stories
- Priorities
- Release 1 and Release 2.

Question3: E-Commerce / Online Shopping App.

User Activities

1. **User Registration & Login**
2. **Search Products**
3. **Product Selection**
4. **Shopping Cart**
5. **Payment**
6. **Order Tracking**

User Tasks

Activity	User Tasks
Registration & Login	Register, Login, Forgot Password
Search Products	Search, Filter, Sort
Product Selection	View Product, View Reviews, Select Size/Color
Shopping Cart	Add to Cart, Change Quantity, Remove Item
Payment	Select Payment Method, Enter Details, Confirm Payment
Order Tracking	View Order, Track Shipment, Cancel Order

User Stories

- As a customer, I want to **register an account** so that I can place orders.
- As a customer, I want to **search for products** so that I can find what I need.
- As a customer, I want to **add products to my cart** so that I can purchase them later.
- As a customer, I want to **make online payment** so that I can complete my purchase.
- As a customer, I want to **track my order** so that I know its delivery status.

Releases

Release 1 – MVP

- Registration/Login
- Search products
- Product details
- Add to cart
- Payment

Release 2

- Reviews
- Wishlist
- Advanced filters
- Order tracking

Create a User Story Map by organizing these requirements.

Question4:

Your team is planning out the next sprint. You've chosen to fill the sprint by taking stories in priority order from the product backlog and stopping when you reach the first story that won't fit in the sprint. Based on following details, which stories should the team commit to for a sprint?

Table 1: Prioritized story with estimated story points and total estimate in hrs of tasks for that story.

Story	Story Points	Total of Tasks Estimates
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Story 1	5	16 hrs
Story 2	8	16 hrs
Story 3	5	24 hrs
Story 4	3	16 hrs
Story 5	13	32 hrs
Story 6	8	26 hrs
Story 7	5	8 hrs
Story 8	8	15 hrs
Story 9	5	12 hrs

Table 2: Capacity of Team members for given sprint

Name	# days available	Hours / day	Capacity (hrs) <i>You compute this</i>
John	3	4-5	
Matt	5	2-3	
Sally	5	4-5	
Ram	5	2-3	

Question5:

Smart Health Clinic is deploying an integrated Clinic Management System that must handle appointments, EMR encounters, lab orders/results, pharmacy dispensing, billing/insurance claims, and reporting while ensuring privacy, role-based access, audit trails, and interoperability. Daily visits range from 600–800 with peak walk-ins.

- Draw a Level 0 Context DFD identifying all external entities
- Provide a Level 1 DFD by decomposing the following major processes:
- Draft an ERD (entities, keys, relationships) for core concepts such as Patient, Provider, Appointment, Encounter, Lab Order, Lab Result, Prescription, Invoice, Claim, Insurance Policy, User/Role, Audit Log.

Question6:

A project size of 400000 LOC is to be developed. Analyze and evaluate the effort and development time for organic, semi-detached & embedded modes of development from the provided values.

(Note: **Organic**: $a_b = 2.4$, $b_b = 1.05$, $c_b = 2.5$, $d_b = .38$; **Semidetached**: $a_b = 3.0$, $b_b = 1.12$, $c_b = 2.5$, $d_b = .35$; **Embedded**: $a_b = 3.6$, $b_b = 1.20$, $c_b = 2.5$, $d_b = .32$)

Question7:

A bank's online loan system must allow applications, verify credit scores, approve loans within 24 hour, and ensure 99.9 % uptime.

- a. List one functional and one non-functional requirement.
- b. How Sprint contributes to the success of the Agile project?

Question8:

A bank has decided to launch a new Online Banking System that will allow customers to manage their accounts, transfer funds, check balances, and more. The system must be highly secure, user-friendly, and scalable. The project needs to be completed in 12 months. The project team must choose between two SDLC methodologies: Waterfall and Agile. Both methods have their strengths, and stakeholders are uncertain about which approach would work best for this high-priority system.

- (i) Compare and contrast the Waterfall and Agile methodologies in the context of developing the Online Banking System.
- (ii) Which methodology would you recommend for this project? Justify your answer with reasons, considering factors like timeline, customer feedback, and regulatory changes.

Question9:

A company is developing digital signal processing software for one of its latest inventions. The estimated size of the software is 4000 lines of code (LOC). The company plans to use the Basic COCOMO model for cost estimation, and the project falls under the embedded category. The model parameters are given as: $a = 3.6$, $b = 1.20$, $c = 2.5$, and $d = 0.32$, the average cost per person-month is ₹50,000. Using this information, calculate the following:

- | | |
|-----------------------|--------------------------|
| (i) Effort Estimation | (iii) Time Estimation |
| (ii) Cost Estimation | (iv) Manpower Estimation |

Question10:

A software project is estimated at 400 FP. A team of five members (one project manager, two senior developers, one junior developer, and one tester) is assigned. Their monthly salaries are ₹90,000, ₹70,000, ₹50,000, and ₹45,000 respectively. The average productivity is 10 FP per person-month. Calculate the total cost of the project.

Question11:

(i) A simple stand - alone software utility is to be developed in 'C' programming by a team of software experts for a computer running Linux and the overall size of this software is estimated to be 20,000 lines of code. Considering $(a, b) = (2.4, 1.05)$ as multiplicative and exponential factor for the basic COCOMO effort estimation equation and $(c, d) = (2.5, 0.38)$ as multiplicative and exponential factor for the basic COCOMO development time estimation equation, approximately how long does the software project take to complete?

(ii) A company needs to develop a strategy for software product development for which it has a choice of two programming languages L1 and L2. The number of lines of code (LOC) developed using L2 is estimated to be twice the LOC developed with L1. The product will have to be maintained for five years. Various parameters for the company are given in the table below.

Parameter	Language L1	Language L2
Man years needed for development	LOC/10000	LOC/10000
Development cost per man year	Rs. 10,00,000	Rs. 7,50,000
Maintenance time	5 years	5 years
Cost of maintenance per year	Rs. 1,00,000	Rs. 50,000

Total cost of the project includes cost of development and maintenance. What is the LOC for L1 for which the cost of the project using L1 is equal to the cost of the project using L2?

Question12:

A software organization is experiencing **frequent project delays, cost overruns, inconsistent development processes, and poor defect management**. Management decides to conduct a CMMI assessment.

As a CMMI assessment team member, explain how you would conduct the assessment. Identify the process areas that should be investigated, the evidence that should be collected, and how the assessment results could be used to improve the organization's processes.

Question13:

Explain Halstead Software Metrics as a technique for software size and complexity estimation.

A development team is analyzing a banking application module containing **25 distinct operators, 40 distinct operands, 150 total operators, and 250 total operands**.

Using Halstead Software Metrics:

- Estimate the **program vocabulary and program length**.
- Calculate the **estimated program length**.
- Calculate the **program volume**.
- Calculate the **program difficulty**.
- Calculate the **programming effort**.
- Estimate the **development time and delivered defects**.
- Interpret the results from a software engineering perspective.

Question14:

A software company has received a project to develop an Online Hospital Management System. The client requirements are well defined, but the management wants to estimate the project size before development.

- Calculate the Unadjusted Function Point (UFP) for the following data:

External Inputs (EI) = 18, External Outputs (EO) = 15, External Inquiries (EQ) = 10,

Internal Logical Files (ILF) = 8 and External Interface Files (EIF) = 4.

Use the weights: EI=4, EO=5, EQ=4, ILF=10, EIF=7.

- Explain how Function Point Analysis assists in software project estimation.

Question15:

A university plans to develop an Online Library Management System where students can search books, issue books, return books and librarians manage the inventory.

- Develop the Level-0 (Context Diagram) for the system.
- Construct the Level-1 Data Flow Diagram (DFD) showing processes, external entities, data stores and data flows.